

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

MACHINE LEARNING FOR UNSTRUCTURED DATA

(CSE (Artificial Intelligence and Machine Learning))

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) What is unstructured data? | 2M |
| (b) What is the use of signature file in text indexing? | 2M |
| (c) Name two application of Text Categorization. | 2M |
| (d) What is machine learning? Name two examples of machine learning algorithms? | 2M |
| (e) What is Information extraction? | 2M |
| (f) What is Anaphora Resolution? Give an example of how does it resolves pronoun. | 2M |
| (g) How does human-centric knowledge discovery paradigm is significant for presentation layer of systems used for data mining? | 2M |
| (h) What is meant by Browsing with respect to text mining systems? | 2M |
| (i) What is Partitioning of Network? | 2M |
| (j) What is Unsupervised Learning for Text? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain document selection method used in text retrieval in detail. 10M
- OR**
- 3 How to use Background Knowledge for Text Mining? Explain the tagging approach for text mining. 10M
- 4 How does usage of unlabelled Data help to Improve Classification? Explain in detail. 10M
- OR**
- 5 What are clustering tasks in text analysis? 10M
- 6 Mr. X is happy someday and angry on other days. We can only observe when he smiles, frowns, laughs, or yells but not his actual emotional state. Let us start on day 1 in the happy state. There can be only one state transition per day. It can be either to happy state or angry state. Draw the Hidden Markov Model for the above. 10M
- OR**
- 7 Given a document collection and a query, describe how a probabilistic model can be used to extract the relevant information from the documents. 10M
- 8 What is Text Visualization? Give the different approaches used for text visualization and explain in detail. 10M
- OR**
- 9 Explain the document explorer system in detail. 10M
- 10 Explain use paths and Cycles in graphs for link analysis in detail. 10M
- OR**
- 11 Give four application of text mining and explain each in detail. 10M

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

MACHINE LEARNING FOR UNSTRUCTURED DATA

(CSE (AI&ML))

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Define Text Mining. | 2M |
| (b) What is the importance of background knowledge in Text Mining? | 2M |
| (c) List the applications of Text Categorization. | 2M |
| (d) Outline the metrics used to evaluate the quality of clustering results. | 2M |
| (e) Define Anaphora Resolution. | 2M |
| (f) Describe the basic principle behind Maximal Entropy Modeling. | 2M |
| (g) Mention the different types of accessing constraints. | 2M |
| (h) What architectural considerations are important when designing visualization tools for text mining? | 2M |
| (i) How does path differ from cycle in graphs? | 2M |
| (j) List the software packages used for Link Analysis. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain the core Text Mining operations and their importance in analyzing unstructured text. 10M
- OR**
- 3 (a) Discuss how task-oriented approaches influence the Text Mining process. 5M
(b) What are text mining query languages, and how do they help in extracting meaningful information from unstructured data. 5M
- 4 (a) Distinguish between Classification and Clustering in Text Mining. 5M
(b) Discuss two clustering algorithms used in Text Mining. How can they be applied to analyze textual data? 5M
- OR**
- 5 (a) Describe the document representation methods used in Text Categorization. 5M
(b) Summarize the Clustering tasks in Text Analysis. 5M
- 6 (a) Illustrate the architecture of Information Extraction systems. 5M
(b) Analyses the applications of HMM in Textual Analysis. 5M
- OR**
- 7 (a) What are inductive algorithms and how do they differ from rule based algorithms in information extraction? 5M
(b) Analyses the applications of CRFs in Textual Analysis. 5M
- 8 (a) How Simple Specification Filters are used at the presentation layer? Give examples. 5M
(b) Discuss common visualization approaches that can be applied in Text Mining to reveal patterns in data. 5M
- OR**
- 9 (a) How does the presentation layer interact with the underlying query language in an Information Extraction system? 5M
(b) Outline the visualization techniques in Link Analysis. 5M
- 10 Distinguish different layout techniques used in Link Analysis with their merits and demerits. 10M
- OR**
- 11 Describe how text mining is used in life sciences research, particularly in mining biological pathway information using systems like Gene Ways. 10M
